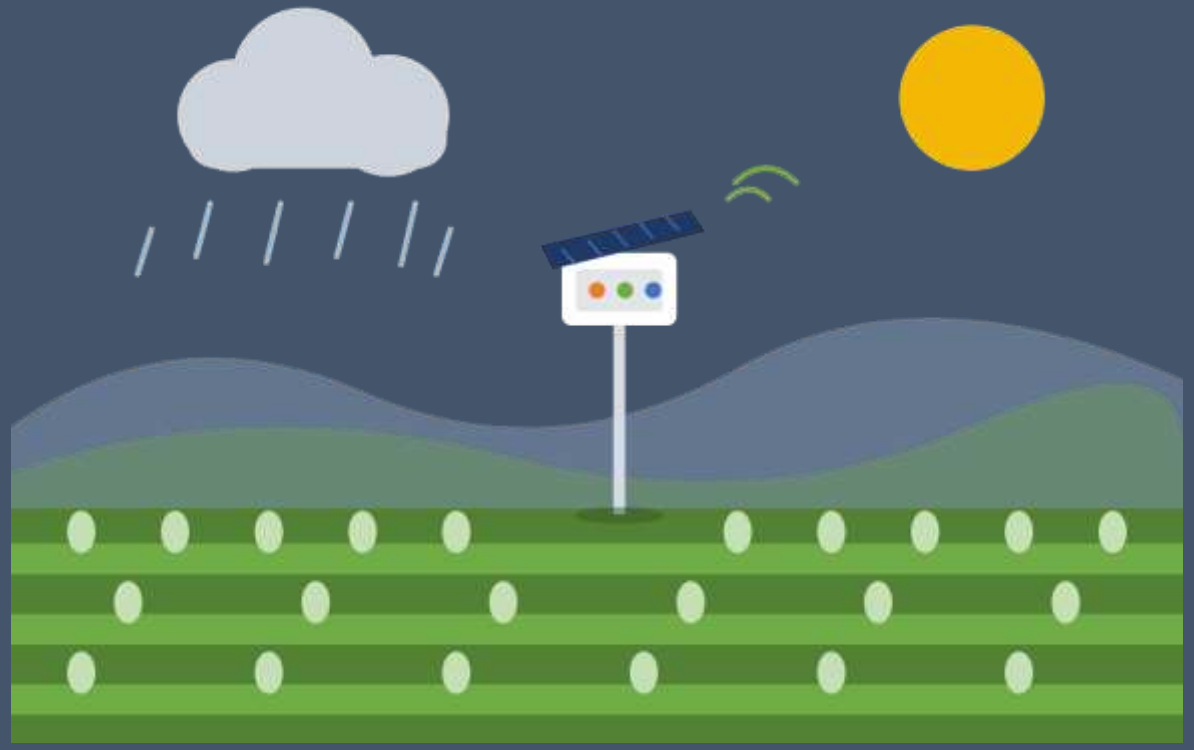
THEME · CLIMATE-SMART AGRICULTURE

# AgriSentinel

## Decentralized proof-of-location for zero-delay crop insurance

Climate risk now strikes one farm at a time, yet insurance still measures it at station or district level. AgriSentinel brings verified, farm-level weather data to the plot, so losses can be proven and paid fast.



**FAO's three pillars of climate-smart agriculture**

| 1 Productivity & income | **2 Adaptation & resilience** | 3 Mitigation |
|---|---|---|

Our focus is pillar 2 (resilience), which protects pillar 1 (income).

# The Invisible Loss

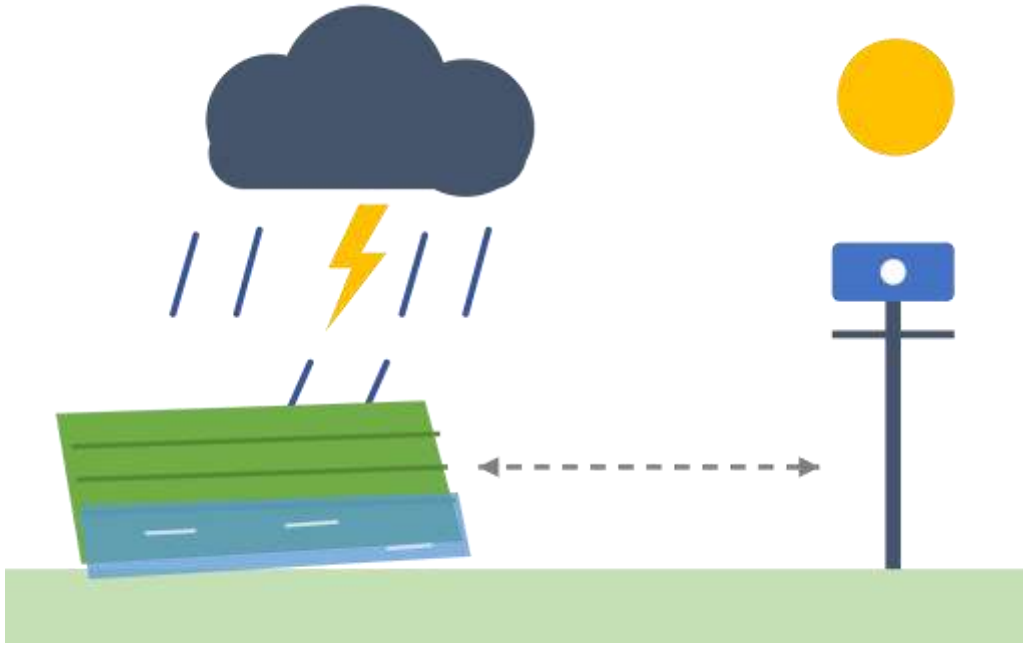
Climate risk is hyper-local. Official weather data is not.

### Basis risk
A micro-burst floods a 2-acre farm, but the weather station 15 km away records zero rain.

### Claim rejected
The crop loss is real, but the storm is not on record, so the insurer denies the claim.

### Evidence gap
Smartphone photos and GPS logs are easy to spoof, so underwriters routinely reject them.

**Basis risk** = the gap between what the station measures and what the farm actually lost.



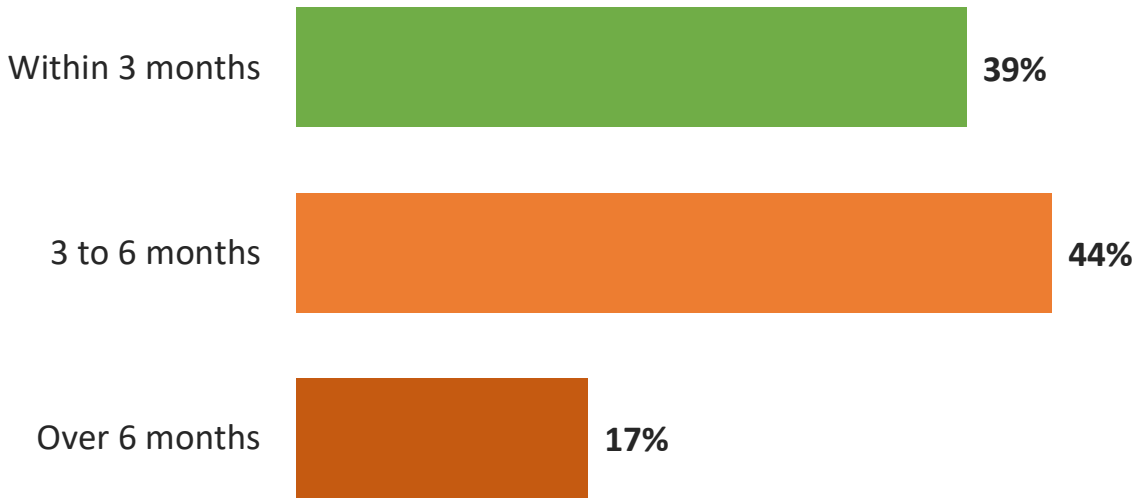
**Our farm: 2 acres, flooded** **Station, 15 km away: 0 mm**

*FAO notes that climate-smart practices are highly location-specific, so resilience tools need farm-level data.*

# Institutions Are Slow and Coarse

Claims take months, and the weather data that could verify losses is too sparse.

**Insured farmers paid, by time taken (Haryana survey, %)**




**Only 39% were paid within the 3-month benchmark.**

## Up to 14 months

Claim delays flagged by a 2025 IIM study, driven largely by manual crop-cutting experiments.

## 12% penalty, weak enforcement

PMFBY guidelines charge insurers 12% a year on late claims. Enforcement varies by state.

## ~800 IMD stations

IMD's national network. Over 20,000 state and private stations exist, but many go unused.

Sources: Haryana PMFBY evaluation (ACR journal); PMFBY revised guidelines; 2025 IIM study as reported in secondary coverage; IMD station inventory via Ministry of Agriculture WINDS note.

THE SOLUTION

# AgriSentinel Edge Node

A solar-powered ESP32 box on each insured plot that measures, signs and reports its own weather.

| | |
|---|---|
| **Power** | Solar panel with battery, off-grid |
| **Controller** | ESP32 reads sensors and signs every packet |
| **Sensors** | BME280 (temp, humidity, pressure), soil moisture probe, tipping-bucket rain gauge |
| **Uplink** | SIM800L: 2G GSM/GPRS, JSON payload |
| **Voice** | Red push-to-talk button and box speaker |

### Measures

Temperature, humidity, pressure, soil moisture and rainfall, all tagged to the plot.

### Detects

Sudden pressure drops and rainfall spikes, checked in real time.

### Matches

Each event is matched to the exact coordinates of insured farms. Target grid: 50 m × 50 m.

TECHNICAL NOVELTY

# Signed Data, Automatic Claims

Proof of loss is signed at the source and verified before any claim uses it.

### 1 Read

Sensors sample the plot on schedule.

### 2 Sign

ECDSA signature. Key stored in read-protected eFuse.

### 3 Send

Signed JSON sent over 2G/GPRS.

### 4 Verify

Backend checks the signature. Forged data is rejected.

### 5 Trigger

Threshold hit (e.g., >80 mm in 2 h). Claim intent to insurer.

### Tamper-evident

Any altered packet fails verification and is never accepted as evidence.

### Skips the manual audit

Qualifying events bypass the 14-month crop-cutting audit. Target settlement: 72 hours.

### Needs a parametric product

Automatic payouts need a parametric product or rider. PMFBY is area-based today.

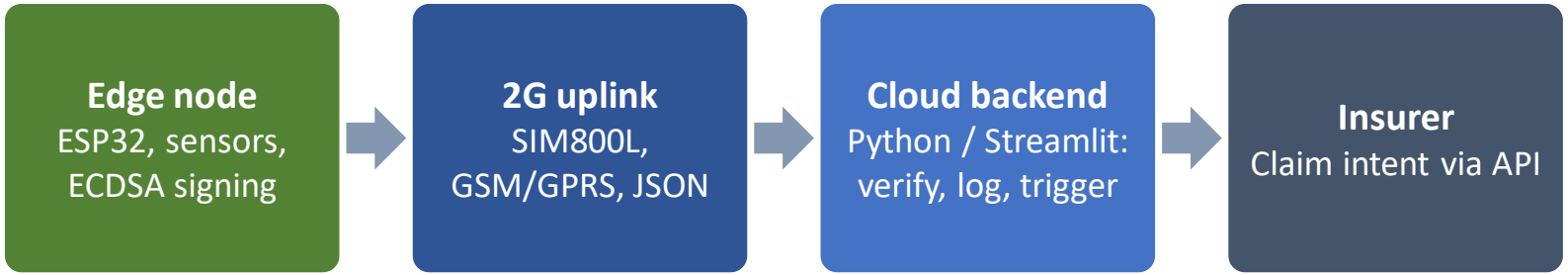
Thresholds are illustrative and must be agreed with the underwriter. Settlement time is a design target, not a measured result.

SYSTEM ARCHITECTURE

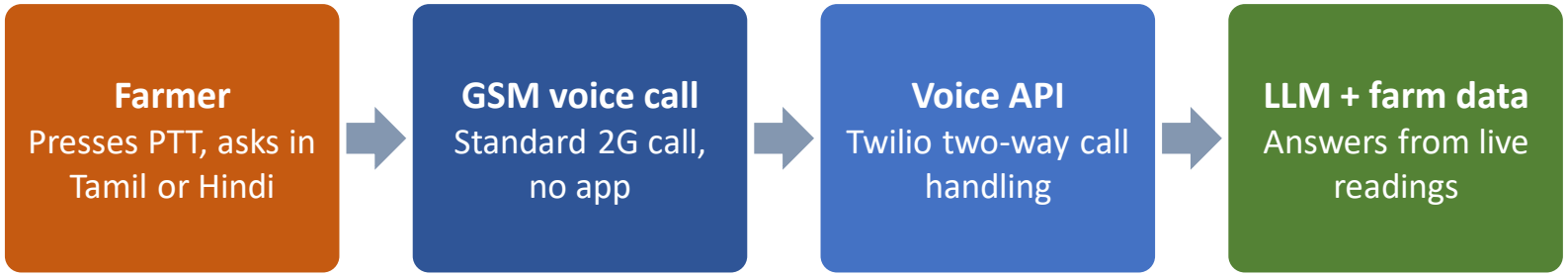
# Built for Low-Bandwidth Networks

Two paths: signed sensor data for claims, and a voice path for advisory that needs no app.

DATA PATH · CLAIMS




VOICE PATH · ADVISORY




## Risks we designed around

**2G may be switched off**
DoT left 2G closure to operators. Plan a 4G Cat-1 modem upgrade.

**Jio SIMs have no 2G**
Pilot on Airtel, Vi or BSNL SIMs instead.

**Classic ESP32 lacks ECDSA hardware**
Use a newer ESP32 variant with an ECDSA peripheral (e.g., C5, H2).

# Beyond the Payout

Insurance pays after the loss. Advisory helps farmers act before it.



## Voice advisory over 2G

The farmer presses the red button and asks, in Tamil or Hindi: “Is it safe to spray fertilizer today?” A regional LLM answers from live farm data, and the reply plays on the box speaker.



## Predictive pest alerts

Blast fungus and spider mites take off under specific temperature and humidity bands. When readings enter a risk band, the node calls the farmer with an early-spray advisory.



## Neighbourhood warning

If one node detects a frost or pest anomaly, the cloud warns neighbouring nodes at once. The whole village gets a shared early-warning shield.

**Why it matters for CSA:** earlier, location-specific advice aims to cut crop losses and reduce blanket pesticide spraying.

# Who Pays, and What We Need

The farmer doesn't pay for the box. The insurer does, because it saves money.

**₹1,200**
target cost per node

**72 hours**
target claim settlement, vs. months today

**50**
pilot nodes in Madurai district

## Why insurers pay

- Fewer fraudulent claims, backed by signed data
- Lower manual crop-cutting costs
- No exposure to 12% late-settlement penalties

## Our ask

Incubation support to deploy our first 50 pilot nodes in Madurai district.

Also seeking an insurer pilot partner, a telecom SIM partner, and an underwriter to define a parametric rider.

**We give farmers an instant voice assistant, give insurers verifiable proof of loss, and end the months-long wait for claim settlement.**